Industrial Placement Report

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with

ChargePoint and ANDEL
Research and Development
ChargePoint: Venture Point Business Park 58, 60 Evans Rd, Liverpool L24 9PB, United Kingdom
Andel: Dodworth Business Park, Andel, Unit 1 Upper Cliffe Rd, South, Dodworth, Barnsley S75 3SP, United Kingdom

From: ChargePoint: 12 / 08 / 2024 – Andel: 24 / 03 / 2025

To: ChargePoint: 10 / 02 / 2025 – Andel: 15 / 08 / 2025

Summary:

I worked with two engineering companies during my placement: I did a 6-month long placement with ChargePoint (based in Speke - Liverpool) and another 5-month long placement with Andel (based in Dodworth - Yorkshire).

ChargePoint is a pharmaceutical containment, aseptic processing and sterile powder transfer specialist. Their main product is the ChargePoint valve, a device that connects two containers and helps transfer fine powder from one container to the other. I worked in the engineering R&D department where I worked on a large variety of tasks, ranging from design to documentation. The main project I was hired to complete was the CPQ project, an initiative to gather all information of all products sold by ChargePoint into one main database. As this project was completed with large advance, I was moved onto other R&D projects.

Andel is a company that specializes in leak detection and prevention, flood defence, wastewater drainage and oil storage, maintenance and compliance. They have a large range of products, but the company originally started with only one product line: BundGuard. BundGuard is a system designed for safe dewatering of bunds, which are containment areas for potential spills or leaks, typically around oil storage tanks. I worked in the R&D / Engineering department where the majority of my tasks were design based. Most of my work was based around the improvement of existing Andel products, adapting them for their new SLS 3D printing machines. The largest project I worked on was the redesign of the BundGuard Sensor.

I learnt a lot during this placement, becoming familiar with two new CAD softwares, learning standard engineering practices such as the use of standards and off-the-shelf components and how to behave in the professional world of engineering. I am grateful for all the guidance I have received throughout this placement by my superiors and for everything these two companies have taught me.

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List of symbols / acronyms

- R&D: Research and Development
- IBC: Intermediate Bulk Container
- FEA: Finite Element Analysis
- ECN: Engineering Change Note

ChargePoint

What is ChargePoint

ChargePoint is a pharmaceutical containment, aseptic processing and sterile powder transfer specialist. The first ChargePoint valve was developed in 1997 with Glaxo Pharmaceuticals (now GlaxoSmithKline) and formed the base of what ChargePoint valves do today. ChargePoint now offers a range of high containment and sterile powder transfer systems along with extraction and decontamination equipment and cleaning, washing and handling accessories. They have expanded their product line by making plastic and sterile single use systems to complement their stainless-steel valves. Their product lines include: PharmaSafe (their baseline products), AseptiSafe (a higher tier), Bio Valves, PuroVaso containers, ChargePoint valve accessories and their ChargeBag containers line.

My Department

The engineering department is split into three teams: the Plastics and Single Use team (focused on the single use, PuroVaso and ChargeBag lines of product), the Contracts Engineering team (focused on order processing and special order requirements, this is where the other engineering students were placed) and the Development Engineering team of which I was a part of.

The Development Engineering focused on R&D and product development as well as any other relevant engineering task. I was hired as an extra engineer to complete the CPQ project and to work on any other lower priority development task that the other engineers did not have enough time to get to. I also helped the other placement students with the orders processing and any other order related tasks. The CAD software that ChargePoint used was Solidworks, so I spent the first couple days learning the basics of Solidworks so that I could start working with it.

My **Projects**

Work Organization

In order to keep track of all of my tasks (ongoing and completed) I created and coded a spreadsheet that I would fill with information about every task I was assigned on. This spreadsheet provided information on what the task was, who it was assigned by (so that I knew who to ask questions to and to whom to send the deliverables), any notes or important aspects of the task, due date and its status. The colour of the row would change depending on what stage the task was at to help the reader quickly identify at what stage the task was. This was very helpful as both my superior and myself could keep track of my workload and what I should focus on as this spreadsheet was accessible to everyone. It also helped for when I would go on leave as anyone could complete an urgent task and had access to all of the task's information.

Task	Type	Size	Priority	Description	Notes	Person / People Responsible	Due Date	Stage	Status
Assesment Report - Lifting Point Valve TEST-0260	Documentation	Medium	Medium	Write an assessment report that looks into whether valves can be lifted by the handles	Report to review lifting points on locking ring that assembly want to use. Looking to consider: - maximum that can be lifted by 4x M6 mounting screws - maximum that can be lifted by locking ring/bayonet - Key risk areas of lifting via handles - Potential design for new bracket to mount to locking ring to fit M10 lifting eye LP NOTES: - Refer to: CTL19PED0002a (P34-36 for bayonet flanges, P53-54 for screw tightening, P23 for pressure forces) Finish section on handle bending moment	Mathew Carmichael	02/10/2024	Other Documention	COMPLETED
MV10228	Part Update	Small	Low	Look over MV10228 with Ian Wadsworth and Tom Wood	SM4134 issue Pin bottoming out on part in assembly Drawing needs clarity on the "Flat bottomed" comment	N/A	N/A	Part	On Hold
CPQ Project	Documentation	Large	High	Gather relevant information about all products and complete spreadsheets with that information		Jack Bullen, Mathew Carmichael	01/02/2025	Other Documention	AWT APP
Combined Valve Assembly	Drawing	Medium	Medium	Make a combined assembly drawing for an Actuated Active Valve and either a: PowerDock, Compensator or RaiseLower	Look at ECN01950	Mathew Carmichael	N/A	Drawing	COMPLETED
PuroGrab Vibratory Attachment	New Product Development	Large	Medium	Create an attachment for the PuroGrab line of products that connects it to a vibratory piston	Project ID: CTL24DEV1595 - PuroVaso Vibratory Collar 00VC-320 - Vibratory Collar with Piston 00BH-320 - Vibratory Collar without Piston PuroGrab: PG-140 PG-240 PG-350 Threaded insert: F-TI-M6-M12-S3.SLDPRT Vibration Piston:https://www.vibtec.com/files/vibtec/industrial_pneumatic_vibrators/linear_pneumatic_vibrators/L1012-2021-Vibtec-FPLF-Pneumatic-Vibrator.pdf (FPLD-25-M) **Check clearance when PuroGrab is on the lifting rig, if there's a problem, make a solution and discuss with Mike Parker (discuss putting PG on backwards).** **Then make Winman products** **RD0569 - RD order containing the PuroGrab Vibratory Collar** **Begin Testing** - Test report: TEST-0273	Mathew Carmichael	N/A	Pre-Testing	ONGOING
Winman Order Tasks	Documentation	Ongoing	Medium	Complete documentation for any orders assigned		Ozkan Tuncer, Mattew Scaife, Mathew Carmichael	N/A	N/A	AWT APP

Figure 1 showing a part of the Tasks spreadsheet

CPQ

ChargePoint had a lot of different databases with different information on all of their products, products are defined as any item ChargePoint has in stock, both the sellable and the non-sellable items. Products can range from a full valve assembly to a bolt they have purchased, having different databases for their products made finding relevant information difficult, so they wanted to gather all the information of every product and make a single database from where all this information could be sourced. I was hired to create this database.

The database was going to be created by filling a number of spreadsheets with the product information and upload the data onto WinMan, their product managing software. Each spreadsheet had the item product codes and a set of fields to fill in. I started filling the database by hand, but then realized that I could use the product codes to fill in some information automatically, that way if any new products were added, some information would already be filled in. By using built in excel functions, I was able to fill in a lot of the cells automatically.

This is an example of how I extracted the information:

Let "10T5-B01-S-05" be a code, 10 indicates the entrance size: 100mm, T5 is the designation of the product, B01 is what type of connection the product has, S is the material used in this product: Stainless Steel, and 05 is the volume of the product: 5 liters. I put this information in a separate general excel sheet that would contain all the characteristics of all the products, what their values are and what code syntax they are associated with. I would then use this spreadsheet to populate the database with the correct information. I

first used the “left”, “right”, “mid” and “textafter” functions to isolate a specific part of the code, for example: textafter(A1, “-“, 3) would output B01. I would then use the “index” and “match” functions to take that output and match it to a specific table in the general data spreadsheet that would contain the correct information. The final output would be the data of the product for that specific category: Volume: 5 liters.

The same principle and functions were used to extract information from other products, the function would be changed to isolate different parts of the code and match it to different parts of the general data spreadsheet. Sometimes the “if”, “iferror” and “find” functions would be used if the code had a different layout. Not all products had most of their relevant information in their codes and so a lot of the database still had to be done by hand, but using excel functions for the codes that did contain information helped speed up the process a lot. A first version of the database was completed four months before the deadline.

The initial difficulty with this task was the repetitiveness as everything was being filled in by hand. Once the possibility of using excel function was explored, the difficulty became learning this new coding language and discovering whether certain things could be done with excel spreadsheets. Nobody in the office had a good enough knowledge of excel to fully help, so most of this project and coding was completed by myself. The excel knowledge I gained from this project helped me code my tasks spreadsheet and will help me in the future with any other excel related task, it also sharpened my research and problem solving skills.

Wash System Mounting Frame

This was my very first proper engineering task, I was tasked with designing a mounting frame that had to support a new experimental product in order for its wash system to be tested. The structure had to support a weight of 100kg, it had to have a specific attachment point at the top and had to securely fix to an IBC. I designed a stainless-steel structure that was made from sheets of metal and stainless-steel extrusion beams. Two off-the-shelf RS Components clips were going to be used to secure the structure to the IBC. I learnt how to look for steel suppliers and how to make a design that used the elements that they sold and how to correctly research for off-the-shelf components. I also applied basic beam loading theory to test whether my design would fail or not, I then double checked my calculations by using FEA software. This structure was going to be made from welded pieces of steel so the main challenge with this project was making a technical drawing that showed the correct welding symbols and how to assemble this structure. I learnt how to use welding symbols and any design considerations that come with making a welded design, such as material warping. This project also highlighted how one must be through with dimensioning a drawing so that the technicians making the part are able to manufacture it. The frame mounted on perfectly to the IBC and was able to hold the full load of the experimental product making this first task successful.

Valve Lifting Test Report

ChargePoint’s bigger valves can weigh over 50kg, if fully equipped, they can reach a maximum weight over 100kg. The assembly and testing team was installing a lifting rig to help move around the heavier valves and they wanted to ensure that the valve handles (used for the manual locking operation) were a secure point from which to lift the valve. I was tasked with writing a test report that did a theoretical analysis of using the handles as lifting points.

I focused on two aspects of the valve for my analysis, the strength of the connection between the handles and the top component of the valve and the strength of the connection between the top component of the valve and the rest of the body. Luckily an analysis was already conducted on the strength of the connection

between the top component of the valve and the rest of the body as that connection would receive load if the valve was sealed and pressurized. I was able to use those results to calculate the maximum load that connection could hold before failure. I then focused on the connection between the bolts and the top valve component. As the bolts were bolted in, I used shear stress analysis to calculate the maximum load that size bolt could use. I assumed that the bolt was holding together two frictionless plates and that the force on those plates was exerted at a right angle to the bolt's orientation, bending moment of the handle was ignored as it was assumed the valve would be strapped as close to the body as possible. I concluded that using the handles to lift the valve was safe as both connections could hold much more load than the heaviest valve configuration.

I was then tasked with creating a handle attachment variant where a lifting eye could be attached. I was given a specific size of eyebolt to use so that the lifting rig could be used, so I performed another analysis to see if the eyebolt would be pulled out by the load of the valve. I assumed the force would be parallel to the direction of the eyebolt and the result was that one eyebolt had a strong enough connection to lift more than the weight of the valve, therefore two eyebolts were more than sufficient to lift the heaviest valve configuration. The lifting eye connector component was later used for a set of vertical handles a customer requested.

Writing this report was very similar to writing university test reports, the only main difficulty was researching and explaining the theory that I used to perform the calculations but for the rest, this was a straightforward project.

Other Design Tasks

These are three examples of the smaller design tasks I was assigned to do. I was tasked with designing a compensator variant (a compensator helps a valve dock more easily to a container) that used steel springs instead of silicone springs. I calculated what spring coefficient was required and selected the appropriate springs. For anti-contamination reasons, the springs had to be covered, so I talked with the ChargeBag engineers to see if we could use a similar material and the same manufacturing methods to produce a cover for the springs. We went over my cover design to make it easier to manufacture and investigated different materials that had the correct properties to use for this cover. I also learned about the correct process for O-ring groove design as the cover had to be sealed. I created test procedures to evaluate whether my design worked but my placement concluded before I could conduct the tests. This project gave me insight on how to collaborate with other engineers from different fields and how to use their knowledge to come up with new solutions.

I was tasked with creating a vibratory attachment for the PuroVaso line of bottles, this was a full new product development task, so I was shown what steps to follow and what documentation to complete. The design was quite simple as a very similar product already existed for the normal ChargeBottle line; the main focus of this project was the documentation. Every step had to be carefully recorded to be able to trace every engineering decision in case it had to be changed. I started writing test procedures for this new product, but I had left the company before the first prototype was made.

I also participated on smaller tasks like a sensor box redesign that halved the size of the sensor box, this project was fun as I set myself the challenge of mounting all sensors on a flat plane instead of having them mounted one on top of the other. I also looked at an old trolley design, I took it apart completely and identified potential improvements that could be made to the design. I assisted another engineer with the creation of a new set of drawings for different electrical and pneumatic plug configurations as the existing products were being replaced.

WinMan Orders Processing and ECNs

The other placement students' main task was with order processing; all orders are logged onto WinMan along with all of the items contained in that order. Their main task was to generate the Engineering Specification that contained all the relevant information about the products along with the necessary safety documents that went with it. The main difficulty of this task was attention to detail as the documentation had to be done correctly, for the rest it was quite monotonous and repetitive.

Engineering Change Notes or ECNs are documents that must be completed when a change to an item is made. Depending on the size of the change, different departments must sign off to approve the change(s). ECNs are a regular part of engineering as each change must be kept track of so that it can be traced back to a date and reason, this helps engineers keep track of changes to products, but this process would sometimes slow progress as departments were not always quick with their approvals.

ANDEL

What is ANDEL

Andel is a Dodworth based company that specializes in leak detection, water sustainability, oil storage maintenance, environmental protection and flood defense. Andel was founded in 1992 and debuted with its mainline product: BundGuard® whose function was to monitor the water and oil levels inside of bunds placed under powerful electrical systems and power grids. Over the years, Andel continued its development of BundGuard®, improving its sensing capabilities and expanding the product range to suit different environments. They also started developing products for other industries such as flood defense and leak protection adding to their available product range. Most of Andel's products are manufactured in house, making design improvement fast and easily implementable.

My Department

I worked in the engineering / R&D department whose main functions are to design and develop new and existing products, and to help customers with technical support and schedule product maintenance. I worked on the product hardware, and my focus was to improve their existing product lines and develop new product hardware. I also created technical drawings and assembly manuals for the manufacturing team to use when assembling the products. The CAD software used by the company is Fusion, it is very similar to Inventor, but I still had to familiarize myself with it before starting to work.

My Projects

Existing Products SLS Manufacturing Redesign

Andel recently had two new SLS 3D printers installed to complement their FDM printers and to permit them to make products with more refined geometry. I was tasked with adapting the existing designs (or redesign completely) for the new SLS printers. The first product I looked at was their point sensor, I made a new CAD model of the enclosure that was fully configurable so that certain dimensions of the enclosure (like the width, length, height, wall thickness and mounting slots locations) could be easily changed. I learnt how to use the configure and parameters functions in Fusion, meaning that I could now make highly configurable design, this was something my old boss at ChargePoint had suggested I learn how to do (using the Solidworks functions) as it could be a very useful skill in the future. One must thing in a very different way when using these tools to make models as a lot of different features are dependent on each other and

so the dimensions must be linked with formulae. It was difficult at first to use these functions, but I can see how this skill can become useful when building future models. I modified the point sensor by adding a snap-on lid to make the opening and closing tool-less and added small groves in the walls to help with the opening of the unit. I used my point sensor model to make the improved AquaeSense® sensor enclosure but was told to keep the screw on lid. I created assembly, general assembly and assembly manual drawings for both the point sensor and AqueSense®. I completely redesigned the Flow-Stop® valve enclosure, reducing its profile and making it easier to assemble and more aesthetically pleasing. Ribs had to be added to the walls of the enclosure as due to its large size, the plastic would sometimes warp after the cooling process was finished. 3D printing makes the prototyping and development process much faster as designs can be quickly iterated upon. I redesigned the Cellar Guard® sensor enclosure maintaining compatibility with the old mounting system and making the product easier to assemble and compatible with the new style sensor probes Andel will implement into their new BundGuard® line. I also used the new 3D printers to print out new jigs and tool mounts to make manufacturing easier and faster.

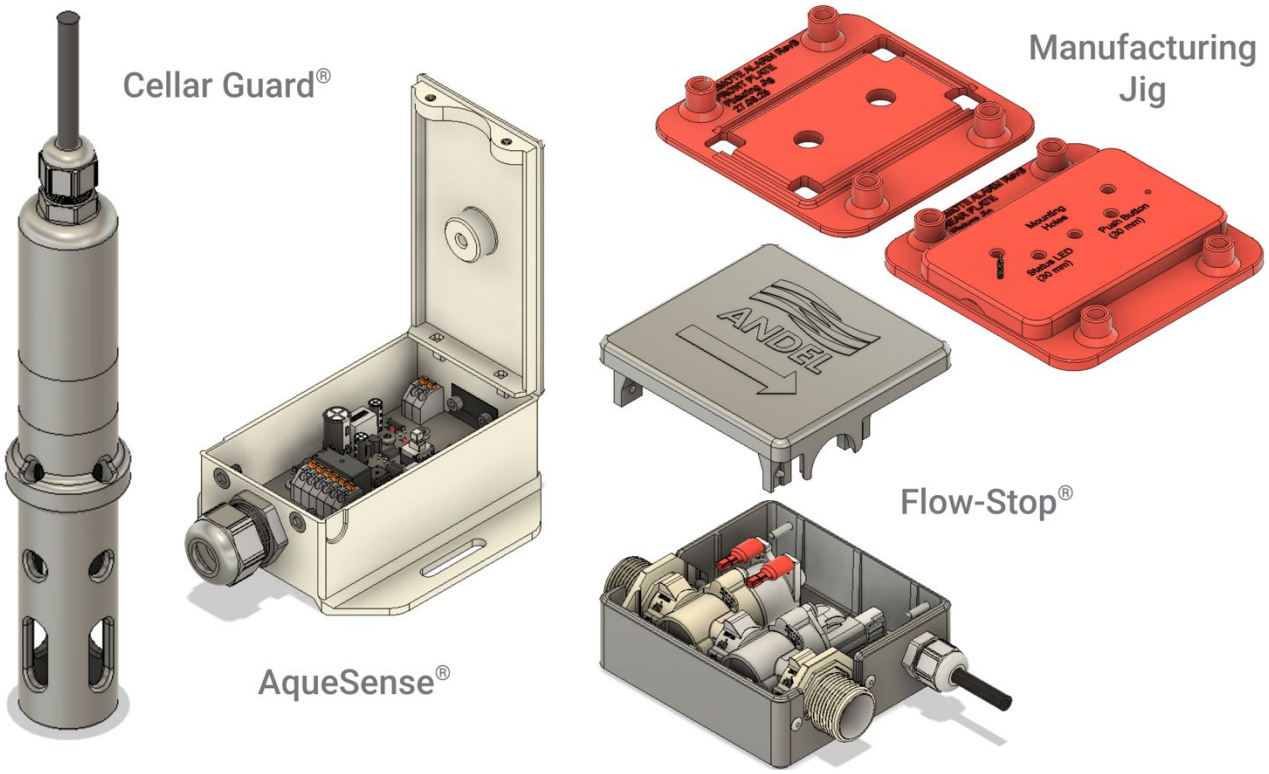


Figure 2 showing a some of the products that were worked on

Water Quality Sensor

Andel started the development of a new water quality sensor with the collaboration of Tecnologico De Monterrey in Guadalajara Mexico. This sensor pack will be placed in a similar type of bollard they already use for other products and will analyse the water that flows into it to determine how clean it is. This product is in its infancy and the design requirements kept changing and evolving. The main challenge of this project was to fit all the desired sensors in the constrained space of the bollard and to design an enclosure that was easily modifiable as the project would keep evolving constantly. The sensor would also need to move up and down according to water level, so a lot of different design options were considered. The bollard used to house it would also need modifications, but the tools available to the supplier are limited, so tools owned by the supplier had to be considered when designing the modifications. I was also shown how to make drawings whose layout and views shown were tailored to machinists to make their job easier. This project was my first experience of early design development and of the difficulties that are

encountered when creating a new product from scratch. A lot of the component dimensions (such as the PCBs and connectors) had to be approximated as the real products has not been developed yet, and the design had to remain flexible to allow changes in dimensions. This project gave me a good understanding of how to organise the initial development of a project and in what order the various components of a product must be designed.

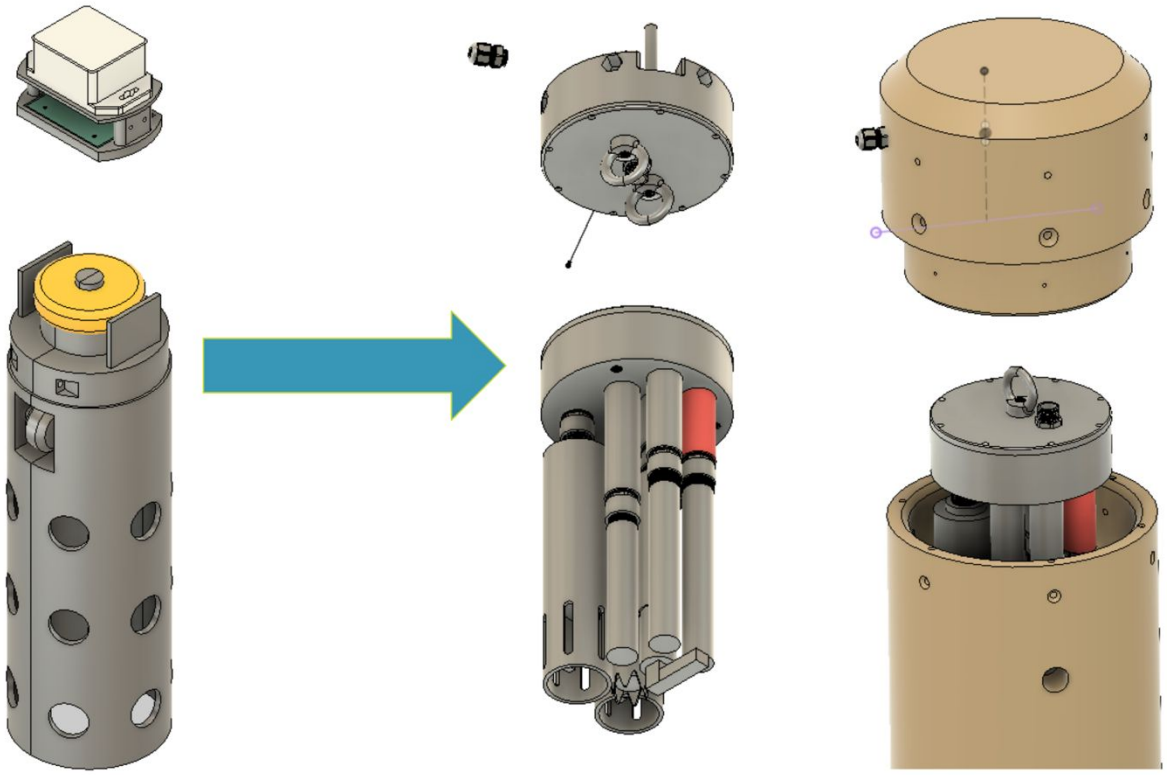


Figure 3 showing the water quality sensor

Gully Sensor

If there is heavy rainfall, sewage systems may sometimes flood, and the flood water would come out through the gullies by the side of the road. Andel wanted to develop a sensor that could alert the user / town before this happened. The sensor had already been selected, and a prototype setup was designed that required a long time to setup and install leading to roads being closed during the install. I re-designed the enclosure to make the installation very quick; to avoid road closures and simple so the installation process did not require many tools and specialized equipment and personnel. The enclosure was made easy to open so that maintenance could be quick. As the enclosure was larger than the build plate of the SLS printers, I had to design it for the FDM printer. This came with a series of design considerations to avoid geometry that required support material. Overhangs had to have 45 degree chamfers underneath for the print to bridge up and some features would have to be supported by super thin walls that would be broken once the enclosure was printed. The print orientation would also have to be considered when designing the enclosure. The sensor was tested on a gully right outside the office and the data received was acceptable to put the item into commercial use.



Figure 4 showing the Flood Alert® Gully Sensor

BundGuard Sensor Redesign

The BundGuard® product is composed of a sensor and a control panel, the sensor is composed from the main body, that houses all the probes, and a base, which serves as a foot for the main body and a place for the pump(s) to rest on. The control panel is a simple enclosure that is mounted outside of the bund and is connected to the sensor(s) via cables. Different BundGuard® models have different sensors which have different numbers of pumps and probes, each model has its own control panel.

Andel had always designed the BundGuard® control panel and sensor enclosures by hand and so they lacked a proper CAD model, so I was tasked with reverse-engineering the existing BundGuard® sensor and control panel models, creating CAD models and drawings and implementing a couple changes that had been planned. I started with the standard sensor model that has four probes and one float, I used the drawings and hand measurements to create an accurate CAD model of the sensor. I used the configuration and parameters functions when making the sensor so that I could quickly make the different variants, the more advanced sensor has an extra float and probe, so it is wider than the original and the mini sensor has the same probes as the standard but is shorter in height. I created a central BundGuard® sensor model and used the parameters function to create all the variants. I did the same with the control panel and made a set of drawings for each sensor and panel to send to the new manufacturer Andel were planning on using.

The changes to the BundGuard® sensor were very minor, and due to how the product line was built, I knew that improvements to the current sensor could be made. I asked the production / manufacturing team if they had any issues with the current BundGuard® design when it came to the assembly and asked the engineering and technical support teams whether there were problems with maintenance or the operation of the unit and made a series of notes on things to improve. I then independently started a complete redesign of the BundGuard® sensor unit. My goals were to make it modular, smaller, easier to assemble and maintain and cheaper to manufacture than the current model.

I shrank the footprint of the main body to a third of the original size and changed the pattern in which the probes were mounted. I mounted the probes on a removable card so that they would not be mounted to the body directly. This would help with maintenance and would allow for only the sensor card to be replaced if a probe broke instead of having to replace the entire unit. The base and pump brackets were now bolted onto the main body instead of welded on, meaning that the same main body could be used for the different BundGuard® sensor models. If a customer wanted to upgrade a BundGuard® sensor they already owned, a whole new unit would have to be sent, with this new model, only the extra parts can be sent as they can be bolted on. The design went through numerous reviews to improve the design ergonomics (mainly the handle size), design for manufacture and implementation of the many different BundGuard® models so that one sensor platform could be used for all the BundGuard® line.

This became the main project I worked on towards the end of my placement and was my first independent product design / improvement project. I learnt a lot from speaking to the suppliers on how to improve drawings and design for manufacture and the design reviews gave me key features to keep in mind

when designing a product. I got to present my new design to the senior managers of Andel during the R&D meetings and developed my skills in being able to present a new design to members of the companies that work outside of the engineering department. This project allowed me to use my technical and creative skills to create a new product, it also helped refine my engineering skills, such as reverse engineering, CAD and drawing skills and product development.

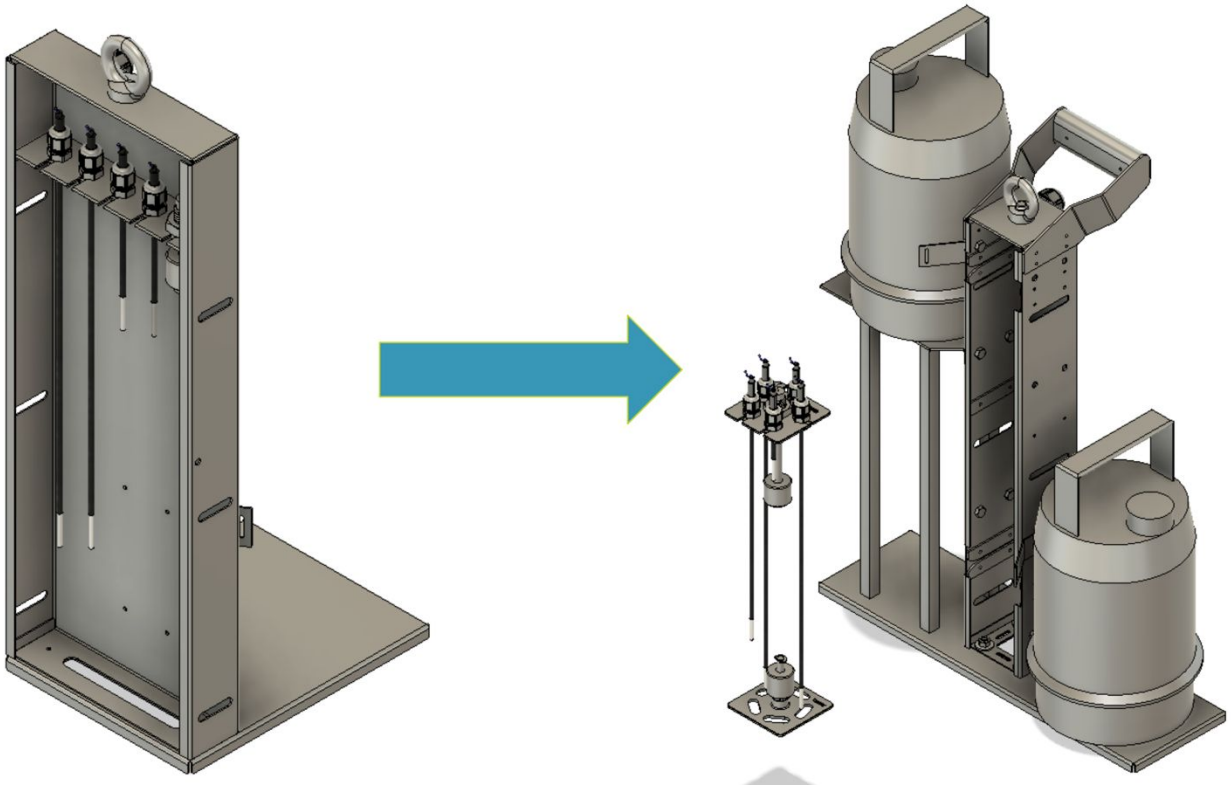


Figure 5 showing the old and new BundGuard® sensors

Conclusion

Reflection on Placement

The work environment at Andel and ChargePoint were very different, ChargePoint's workflow was very focused on formalities and procedures whilst Andel's was more casual and free flowing. ChargePoint worked alongside the pharmaceutical industry and this meant that everything had to be very well documented, whilst Andel did not face the same documentation requirements. The different sizes of the two companies also influenced the work environment as smaller companies tend to be more free flowing. I found that I thrived more in Andel's environment as there weren't as many procedures to learn and found that the more relaxed work environment was more welcoming. But my view of ChargePoint's environment might have been influenced by the little training on procedures that I received and not knowing the proper way to do things.

I found that the design tasks were my favorite part of the placement, going forward I will probably look for more design-oriented jobs in the field of engineering. The most challenging was the documentation aspect, but that again may be due to the lack of training or the complexity of those tasks.

Acknowledgements

I would like to thank the members of ChargePoint's engineering team for the great experience of working with them and Jack Bullen for his guidance and teaching during my placement. I would like to thank the members of Andel for the warm welcome I received when I joined and the engineering team, specifically Joe Cullingford and Harry Merckel, for the instructive and fun experience of working with them. Thank you for having helped me develop as an engineer and as a person and thank you for having made my placement experience fun and a fond memory.

Conclusion

I found this placement experience to be very fulfilling and an essential part of my education in the field of engineering as learning about engineering is very different to being an engineer. University teaches you the theory behind concepts used in engineering and some of the practical engineering skills, but most of the skills that I used during my placement, I learnt on the job. I would like to see the more practical aspects of engineering to be covered a little more in university but ultimately the placement year and summer internships help cover that side of training. I would like to again thank my superiors and companies who gave me the opportunity to work with them during this placement year and for all the training and experiences I gained along the way.